Smart Form with Employee Request–Approval Workflow

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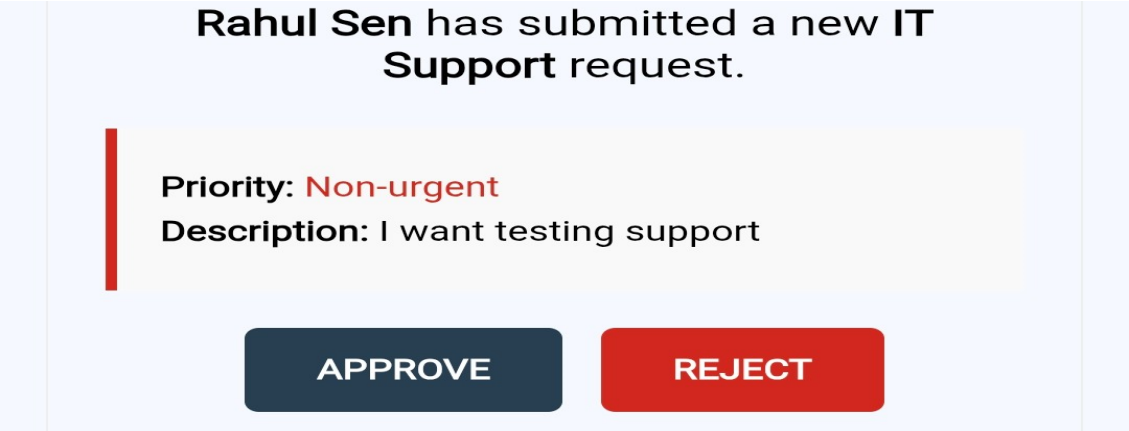
Problem Statement

- **Inefficient Manual Processes** Traditional approval methods are slow, time-consuming, and prone to delays due to manual handling.
- **Lack of Visibility and Communication** Absence of proper tracking and communication leads to poor transparency between employees and managers.

Introduction

- The Smart Request Approval System is a workflow solution that automates employee request submission and approval processes using Google tools.
- It streamlines communication by enabling quick decision-making through interactive emails.
- The system ensures transparent and real-time handling of organizational requests.



Objectives

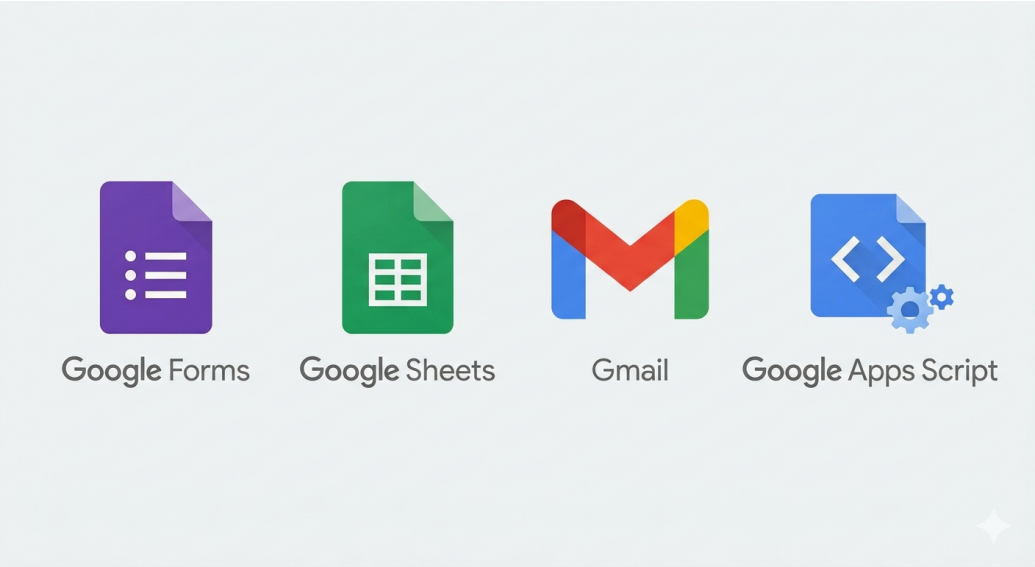
- **Automate Request Handling** Eliminate manual processes by enabling employees to submit requests digitally through a structured form, ensuring faster and error-free data collection.
- **Streamline Approval Workflow** Provide a seamless approval mechanism through interactive emails, allowing managers to approve or reject requests with a single click.
- **Real-Time Notifications** Ensure timely communication by sending automated email updates to both managers and employees at each stage of the workflow.
- **Centralized Data Management** Maintain all request data, statuses, and timestamps in a single Google Sheet for easy tracking and auditing.
- **Improve Transparency and Efficiency** Enhance visibility into request status while reducing delays and improving overall organizational productivity.

System Overview

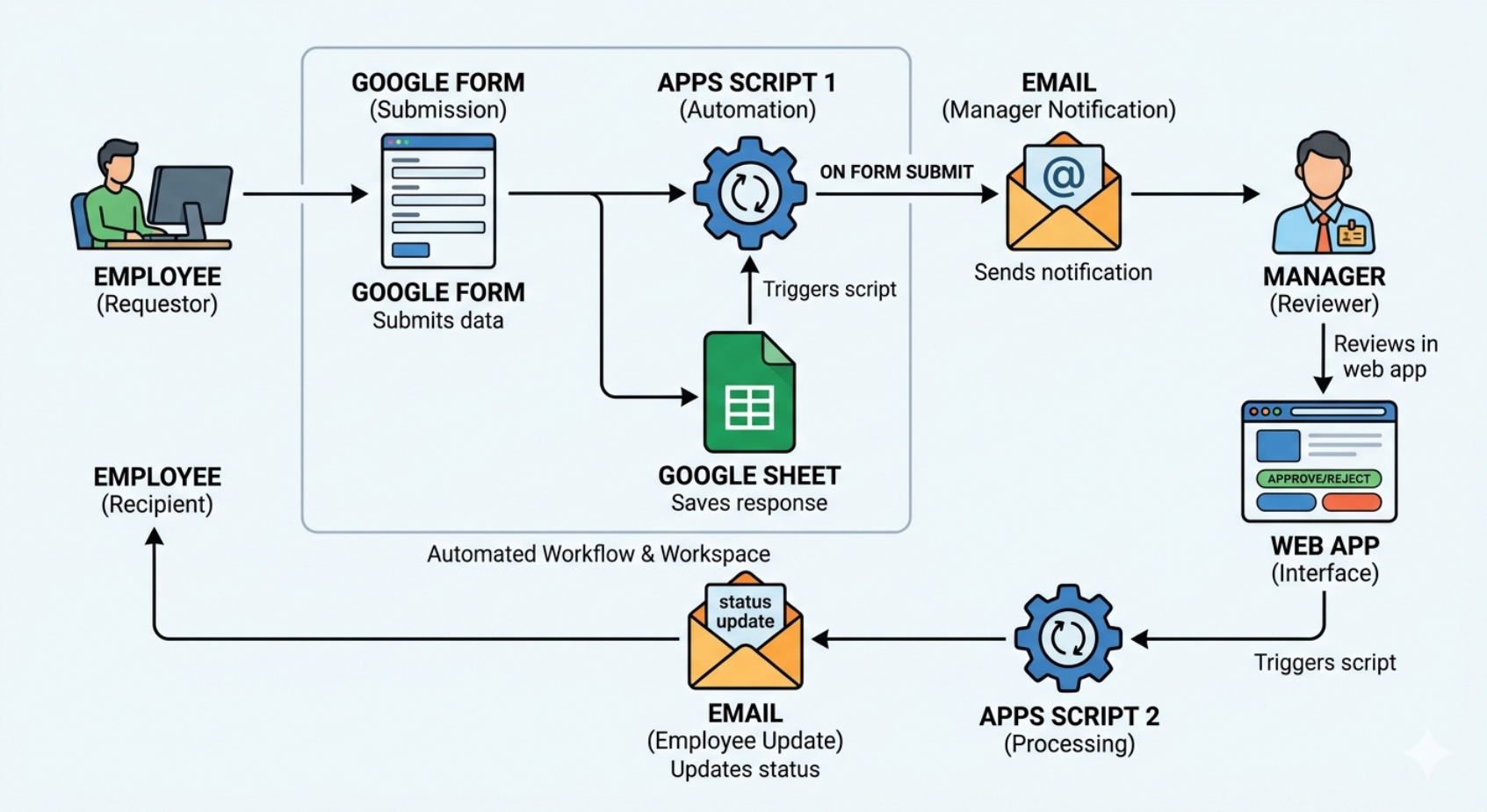
- **Serverless Workflow System** Eliminates the need for dedicated infrastructure by leveraging Google Workspace services, making the system scalable and cost-effective.
- **Event-Driven Architecture** The workflow is triggered automatically on events such as form submission and approval actions, ensuring real-time processing without manual intervention.
- **Central Database: Google Sheets** Acts as a single source of truth to store all request data, track statuses, and maintain timestamps for auditing and monitoring.
- **Backend Logic: Apps Script** Handles data processing, dynamic routing, email notifications, and approval actions through automated scripts.

Technologies Used

- Google Forms
- Google Sheets
- Gmail
- Google Apps Script (Web App Service)



Architecture Diagram



Google Form Design

- **Step 1: Basic Information**
  - Requester Name (Full name, minimum two words)
  - Manager's Email (Valid email format)
  - Request Type: Leave, Access, IT Support, Travel, Budget, Other
- **Step 2: Leave Details (Conditional)**
  - Leave Start Date
  - Leave End Date
- **Step 3: Request Details**
  - Priority: Urgent / Non-Urgent
  - Description (Max 999 characters)
  - Supporting Attachment (Optional, max 10MB)
- **Step 4: Submission**
  - Option to receive response copy
  - Review and submit the form

Google Apps Script Logic

- **Bound Script Integration** The Apps Script is container-bound to the Google Sheets response file, allowing direct access to form data and seamless automation.
- **Event Trigger: onFormSubmit** An installable trigger executes automatically whenever a new form response is submitted, initiating the workflow.
- **Data Extraction** The script retrieves user inputs using the `e.values` object and processes request data, determines request type and priority, and performs validation where required.
- **Web App Handling using** `doGet(e)` When a manager clicks APPROVE/REJECT, the web app is invoked; the `doGet(e)` function reads URL parameters (action, id) and updates the request status in Google Sheets.

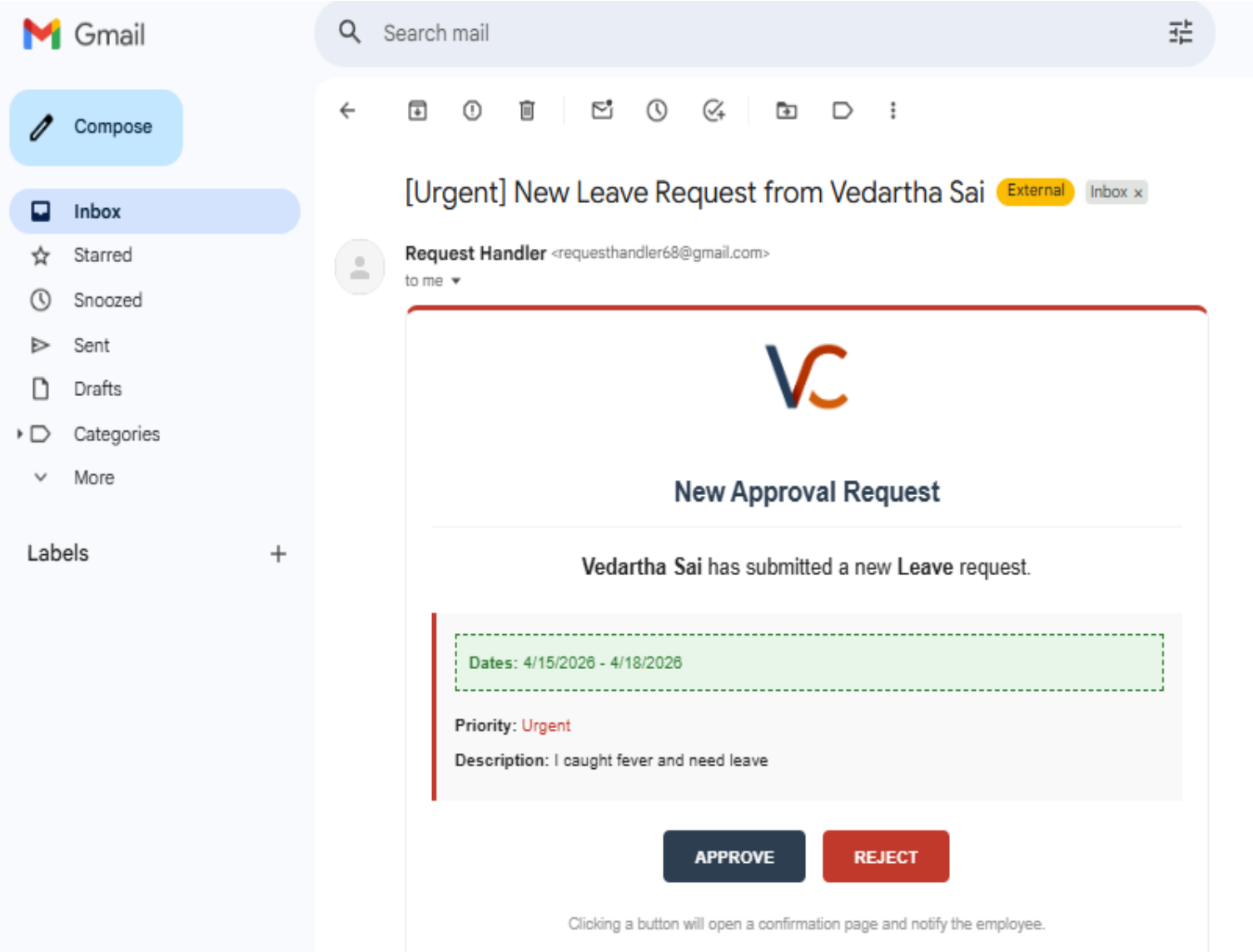
Email Notification System

- **Approval Request Emails to Managers** Automatically sends structured emails containing request details, along with interactive APPROVE and REJECT buttons for quick action.
- **Decision Notifications to Employees** Once a request is processed, the system sends automated emails informing employees of approval or rejection along with the decision timestamp.
- **Interactive Email Experience** Action buttons are linked to the web app, enabling managers to respond directly without accessing spreadsheets.
- **HTML-Based Email Templates** Uses `HtmlService.createTemplateFromFile()` to generate professional, well-formatted emails with clear layout and styling.

Web Application

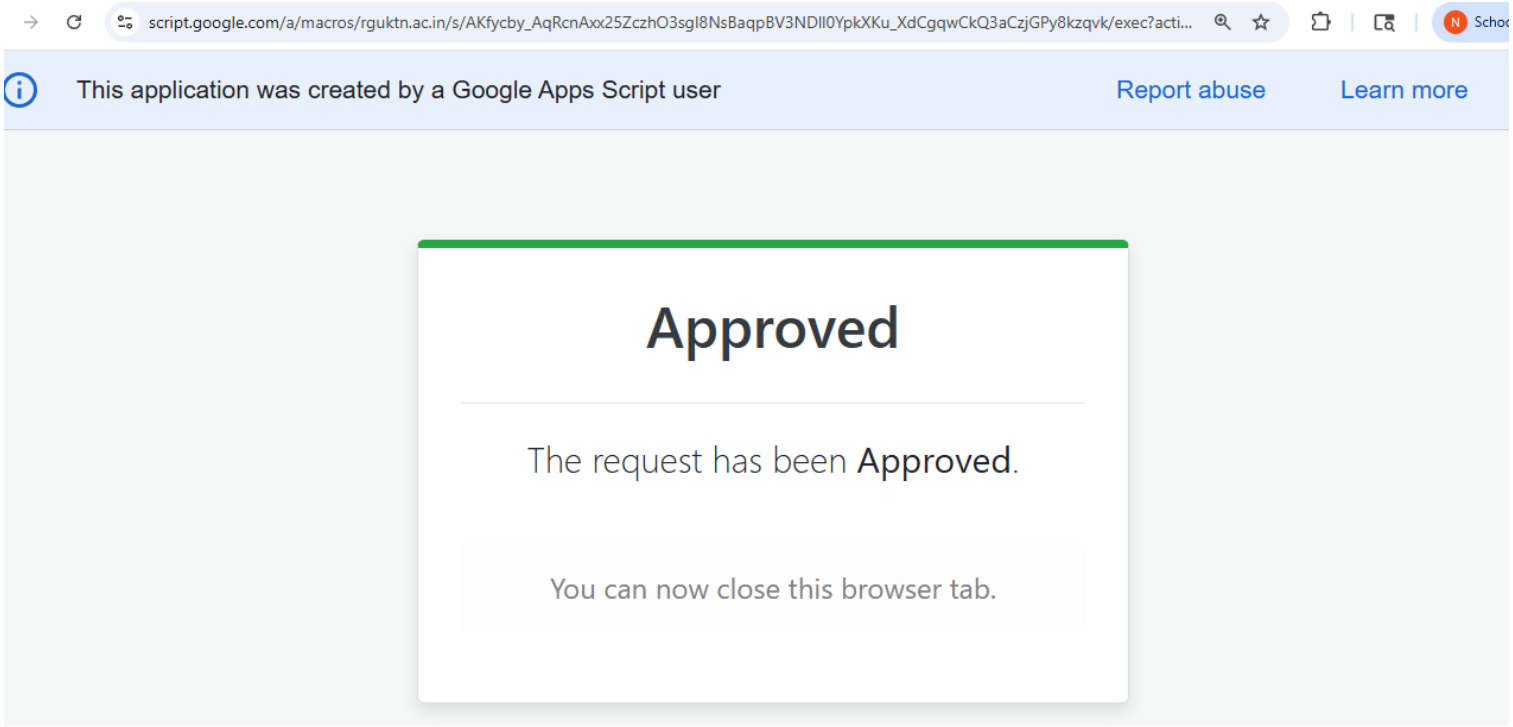
- **Approval via Web Application** APPROVE and REJECT buttons in the email redirect managers to a deployed web application to confirm their decision.
- **Request Processing using** `doGet(e)` The web app's `doGet(e)` function captures URL parameters such as action and request ID, and updates the corresponding record in Google Sheets.
- **Duplicate Action Handling** Before updating, the system checks the sheet to determine if the request has already been processed.
- **Prevention of Multiple Responses** If the same request is accessed again, the web app displays an **"Already Processed"** message, preventing duplicate approvals or changes.

Snaphots



Snapshot 1: Approval Mail

Snaphots



Snapshot 2: Web Application

Snaphots

Request APPROVED

Hello **Vedartha Sai**,

The status of your Leave request has been updated.

Request Type:	Leave
Status:	Approved
Decision Date:	4/14/2026
Approved By:	n201101@rguktn.ac.in

If you have questions about this decision, please contact your manager directly.

Automated Notification System

Do not reply to this email.

Snapshot 3: Confirmation mail

Future Enhancements

- **Multi-level Approval Workflow** Support hierarchical approvals involving multiple managers.
- **Analytics Dashboard** Provide insights and reports on request trends and performance.
- **Extended Request Types** Add more request categories with advanced conditional input fields.
- **AI-Based Email Generation** Automatically generate intelligent reason summaries from form data.
- **Integration with HR Systems** Connect with existing HR platforms for seamless data synchronization.

Conclusion

- **Efficient and Automated Approval System** Provides a streamlined solution for handling employee requests with minimal manual intervention.
- **Improves Productivity and Workflow** Reduces delays and enhances coordination between employees and managers.
- **Demonstrates Power of Serverless Architecture** Showcases how Google Workspace tools can be leveraged to build scalable and cost-effective solutions.

Links & References

- Google Forms Documentation: https://developers.google.com/workspace/forms
- Google Apps Script Documentation: https://developers.google.com/apps-script
- Employee Request Form Link: Form Link
- GitHub Repository: GitHub Repo
- Apps Script Project: Apps Script Project

Thank You

Thank You!